

POLITECNICO DI TORINO



Cybersecurity and National Defence Course Project

Protection of Space-Based Assets from Cyber Threats

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Abstract

Satellites and other space-based systems play an essential role in modern society, supporting communication, navigation, weather forecasting, banking operations and military activities. As these infrastructures become increasingly connected to digital networks, commercial components, cloud-based ground segments and standardized communication protocols, they also become more exposed to cyber threats.

This report analyzes how space-based assets can be targeted through cyber means and how such attacks may affect national security and civilian life. The *first part* examines the technical attack surface of space systems, focusing on environmental constraints: SWaP-C, radio-frequency communication links, onboard buses, AOCS sensors and electrical power subsystem threats. The analysis shows that cyber compromise in space can produce physical consequences such as loss of attitude control, battery degradation, thermal damage or even national service disruption.

The *second part* connects these technical vulnerabilities to terrestrial impact and to risk management. A Python-based cyber risk prioritization and treatment tool is used to model representative scenarios, compute residual risk and evaluate the effect of recommended security controls. Finally, the report discusses regulatory mitigations, with particular attention to the EU GDPR and the Italian National Cybersecurity Perimeter. The objective is to show that protecting space infrastructure requires an integrated approach combining engineering controls, risk assessment and governance.

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Part I

Architectural & Technical Analysis

Chapter 1

Satellite Architecture and Environmental Constraints

To comprehend the unique challenges of satellite cybersecurity, it is necessary to understand the environmental realities and architectural limitations that govern the engineering of space vehicles. The space domain imposes severe restrictions that directly conflict with the computational requirements of modern defensive software suites. These constraints are not background engineering details. They define the boundary conditions under which any cybersecurity mechanism must operate: the available processing power, the energy budget, the orbital environment and the expected mission lifetime all determine which measures can be implemented onboard.

1.1 Types of Satellites and Orbital Mechanics

Modern space infrastructure is diverse, ranging from massive geostationary platforms to tiny constellations in Low Earth Orbit (LEO). Understanding the size and orbital mechanics of these assets is the first step in analyzing their cyber-attack surface.

Subcategory	Mass (kg)	Orbits	Typical Applications and Characteristics
Minisatellite	100–500	LEO, MEO, GEO	Complex scientific missions, high-resolution Earth observation, and telecommunication.
Microsatellite	10–100	LEO, HEO	Remote sensing, atmospheric research, and foundational platforms for broadband.
Nanosatellite	1–10	LEO	Dominated by the standard CubeSat format. Heavily utilized for technology validation.
Picosatellite	0.1–1	LEO	Ultra-compact formats utilized to form distributed networks and stress test new materials.

Table 1.1: Classification of Satellites by Mass and Orbit

This diversity is highly relevant from a security perspective because satellite class, orbital regime, and mission profile shape the attack surface in different ways. Large GEO platforms usually support long-lived, high-value services and complex ground infrastructures, whereas proliferated LEO constellations significantly increase the number of nodes, links, handovers, and third-party interfaces that must be protected.

Critical terrestrial operations rely on specific orbital layers. Global banking operations and timing systems rely entirely on GPS networks operating in Medium Earth Orbit (MEO). Military networks leverage massive platforms in Geostationary Earth Orbit (GEO)¹ alongside newer, highly proliferated tracking layers in LEO. As satellites decrease in size to fit within these mega-constellations, their computational resources become more limited, creating a severe new barrier to cybersecurity integration.

1.2 The SWaP-C Paradigm and Cryptographic Overhead

The design philosophy of all spacecraft is bounded by the strict constraints of SWaP-C (Size, Weight, Power, and Cost). Launching mass into orbit remains one of the most expensive technological endeavors in human history. Consequently, aerospace engineers are driven to miniaturize components and optimize power consumption to the absolute limit. A standard 1U CubeSat leaves no electrical *breathing room* for auxiliary systems that do not directly contribute to the primary mission payload.

These severe constraints present a fundamental barrier to implementing robust cybersecurity measures. Modern cryptographic algorithms, such as the Advanced Encryption Standard (AES) or the Rivest-Shamir-Adleman (RSA), are computationally intensive. For a satellite's On-Board Computer (OBC), which is often an underpowered microcontroller, the overhead of encryption can become significant.²

Furthermore, aerospace systems execute Guidance, Navigation and Control (GNC) algorithms within strict, microsecond-level timing windows. If a cryptographic operation introduces unpredictable latency or causes a cache miss, it could delay the execution of a critical command to a reaction wheel or a chemical thruster, resulting in an immediate and unrecoverable destabilization of the spacecraft's attitude.

¹GEO satellites rotate synchronously with the Earth, appearing stationary, whereas LEO satellites move rapidly relative to the Earth's surface due to their lower altitude.

²Transitioning to asymmetric encryption introduces significant latency and high CPU utilization, drastically reducing the effective throughput of downlinked scientific telemetry and payload data.

1.3 Environmental Radiation and Hardware Security Limitations

Beyond physical limitations, the space environment is characterized by intense ionizing radiation that alters and degrades electronic components. Satellites in LEO pass through regions such as the South Atlantic Anomaly, while higher orbits are subjected to the trapped proton and electron fluxes of the *Van Allen* radiation belts.

This interaction with semiconductor electronics manifests in two failure categories:

- **Total Ionizing Dose (TID):** This refers to the cumulative absorption of ionizing radiation over the operational lifespan. Trapped charges cause a gradual shift in threshold voltages, leading to complete failure of the microcircuit.
- **Single Event Effects (SEE):** These are anomalies caused by a single high-energy particle. The most common is a Single Event Upset (SEU), a transient fault where a particle deposits enough charge to flip a memory cell.

In terrestrial enterprise networks, the foundation of a **Zero Trust architecture** relies heavily on Hardware Security Modules (HSMs) to securely manage critical cryptographic keys. However, deploying commercial HSMs into space is highly problematic. If a high-energy particle induces an SEU within the secure enclave storing an AES key, the key is permanently corrupted, instantly severing the encrypted command link with the ground station. To counteract these effects, the industry utilizes Radiation-Hardened By Design (RHBD) components, but space-grade processors consistently lag several generations behind modern commercial technology.

The consequence is a persistent engineering trade-off: spacecraft must preserve reliability and deterministic behaviour while operating with severely limited computational margins. This trade-off directly affects the protection of communication links, since authentication, encryption, and secure key management must be implemented without compromising mission timing, power availability, or fault tolerance.

Chapter 2

Communication Networks and Threat Modeling

The operational continuity of any space mission depends on uplink and downlink Radio Frequency (RF) communication links. To comprehend the cyber-physical threat surface, it is mandatory to dissect the physical RF front-end and the mathematical encapsulation protocols required to interface with the spacecraft's internal systems.

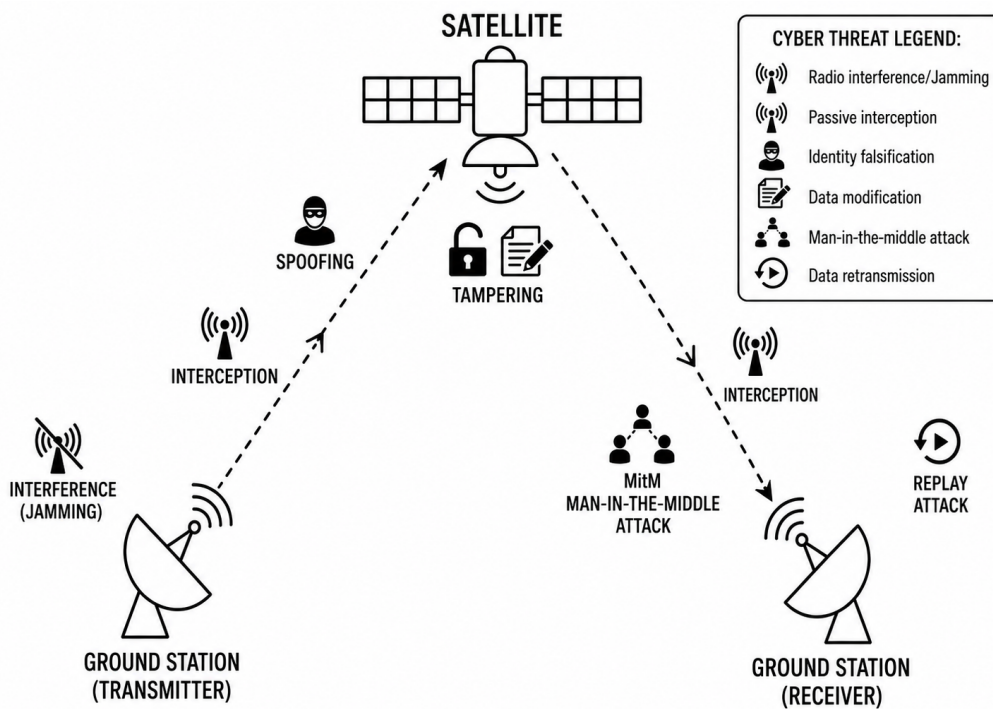


Figure 2.1: Ground-to-satellite communication architecture and main cyber threats affecting the vulnerable uplink, downlink, and ground segment mission infrastructure.

This diagram highlights how the attack surface is not limited to the spacecraft, but includes uplink, downlink, and ground segment. Main vectors include jamming, interception, spoofing, tampering, man-in-the-middle attacks and replay attacks

2.1 RF Hardware Architecture and Payload Dynamics

The physical layer is governed by electromagnetics and severe power constraints. The terrestrial anchor utilizes a highly specialized RF Front-End. Because ground stations operate in full-duplex, mathematically tuned diplexers isolate the kilowatts of energy generated by the High Power Amplifiers from the highly sensitive Low Noise Amplifiers, preventing the station from electrically blinding its reception circuitry.

Upon reaching the space segment, payloads fall into two engineering categories:

- **Transparent Payloads (Bent-Pipe):** The satellite functions as an analog RF mirror. It receives uplink, translates the carrier frequency, amplifies the signal and retransmits it. Because there is zero digital demodulation, it is immune to standard software malware, but extremely vulnerable to physical RF jamming.
- **Regenerative Payloads (On-Board Processing):** The satellite acts as an independent network node. The RF signal is demodulated to baseband and routed. While resilient to analog noise jamming, complex Commercial Off-The-Shelf (COTS) Linux kernels required for routing introduce severe vulnerabilities, including Remote Code Execution.

In both cases, the RF signal becomes operationally meaningful once it is transformed into structured command or telemetry data. This is why the protocol stack is as important as antenna hardware: a malicious transmission must not only reach the spacecraft, but also be formatted in a way that onboard systems can parse and route.

2.2 The Protocol Encapsulation Pipeline

A cyber-adversary cannot transmit raw binary code to compromise a space vehicle. The malicious payload must adhere to the deeply nested encapsulation architectures standardized by the Consultative Committee for Space Data Systems (CCSDS).

At the network layer, commands are encapsulated into the Space Packet Protocol. The critical vulnerability resides in the Application Process Identifier (APID), an 11-bit routing mechanism. An attacker crafting a malicious payload must maliciously spoof the highly specific target subsystem's APID (e.g., targeting the propulsion system) to ensure the payload detonates in the correct internal hardware module.

To traverse the physical RF link, these packets are multiplexed into Telecommand Transfer Frames. To survive thermodynamic noise, the frames undergo Forward Error Correction and are formatted into a Communications Link Transmission Unit (CLTU).¹ Finally, the bitstream is mathematically XORed with a pseudo-random sequence to prevent the receiver from losing timing synchronization. An attacker utilizing a Software-Defined Radio must perfectly replicate this mathematical chain in real-time to execute a successful command injection. This encapsulation chain therefore acts as a barrier and an attack surface. It raises the technical difficulty of command injection, but once an adversary is able to reproduce, replay, or disrupt parts of the chain, the same mechanisms that normally guarantee reliable spacecraft communication can be abused to deliver malicious commands or degrade the link.

2.3 Link-Layer Threat Vectors and GSaaS Vulnerabilities

Adversaries exploit architectural vulnerabilities to attack satellite communication links across multiple vectors. One of the most common methods is jamming, broadcasting high-power noise within the same frequency band used by the transmitter and receiver, severely degrading the communication link's quality. A more sophisticated variation of this attack targets the Adaptive Coding and Modulation (ACM) feedback loop. By manipulating this loop, an attacker can force the gateway to adopt a fragile modulation scheme, triggering a catastrophic Frame Error Rate (FER).

Another significant threat is eavesdropping, the covert acquisition and surveillance of private conversations, electronic messages and data transmissions without consent. While Telemetry, Tracking, and Command (TT&C) links are crucial for a satellite's survival, commercial broadband and Earth observation constellations rely heavily on DVB-S2/S2X standards utilizing Generic Stream Encapsulation (GSE). However, severe and systemic vulnerabilities exist because the industry frequently fails to implement robust encryption protocols like IPsec or TLS. This oversight allows attackers using Commercial Off-The-Shelf (COTS) Software-Defined Radios (SDRs) to perform passive eavesdropping over massive geographic areas. Consequently, they can extract cleartext IPv4 traffic, exposing highly sensitive data such as Voice over IP (VoIP) SIP signaling, caller metadata, and precise terrestrial geolocations.

¹A CLTU is the base physical layer data structure that allows synchronization and error correction over the space-to-ground link.

The lack of robust security exposes command links to critical attacks. If a satellite’s TT&C uplink lacks cryptographic authentication, a common vulnerability due to Size, Weight, Power, and Cost (SWaP-C) constraints of legacy On-Board Computers (OBCs), it becomes susceptible to replay attacks. An attacker can intercept and later replay analog RF waveforms, forcing the satellite to re-execute obsolete but valid commands. Even worse, if an adversary synthesizes a valid Command Link Transmission Unit (CLTU) containing malicious flight software directives, the impact can be catastrophic, leading to total loss of control.

Finally, satellite ground stations rely heavily on Commercial Off-The-Shelf (COTS) hardware integrated with complex custom software, making routine security patching dangerously slow. This leaves a vast attack surface, spanning remote antennas, cloud servers, and third-party vendors, exposed to known IT exploits. The ground segment itself is a prime target for injection attacks. A real example occurred in 2022, when the “AcidRain” wiper malware infiltrated the Viasat KA-SAT management network. This malware was used to remotely break down thousands of terrestrial modems, effectively severing vital satellite communications throughout eastern Europe.

Furthermore, the decentralization of the ground segment toward the Ground Station as a Service (GSaaS) paradigm² allows to route telemetry and telecommands between distributed antennas, relying on the Space Link Extension (SLE) protocol. SLE historically lacks protocol-layer encryption, exchanging highly structured data utilizing cleartext. If an adversary gains a foothold within the ground segment’s network, they can seamlessly route malicious CLTUs directly into the ground station’s amplifier.

²GSaaS allows virtualized Mission Control Centers to lease generic antenna time via cloud APIs, reducing costs but expanding the attack surface.

Chapter 3

On-Board Subsystem

Vulnerabilities and Exploitation

To execute a successful cyberattack against a space vehicle, an adversary must navigate a highly complex, interdependent array of embedded systems. A thorough analysis reveals that the most critical vulnerabilities reside deep within the engineering subsystems of the satellite itself. The communication link can be understood as the entry channel, but the onboard subsystems determine the physical consequences of a successful compromise. Once malicious commands, corrupted telemetry, or falsified sensor data reach the spacecraft bus, the attack is no longer purely informational: it can alter attitude, power availability, thermal balance and mission autonomy.

3.1 Attitude and Orbit Control System (AOCS) Threats

The Attitude and Orbit Control System (AOCS) is responsible for measuring the spacecraft's orientation and generating the torques to adjust it. Because the AOCS relies on a closed-loop, automated architecture that constantly reacts to analog sensor inputs, it is susceptible to sophisticated, physics-based spoofing attacks.

The determination phase relies on specialized sensors. Star trackers capture images of the star field, Sun sensors compute an illumination vector, and Inertial Measurement Units (IMUs) measure angular velocities. According to detailed technical analyses, these sensors can be deliberately spoofed using analog signal injection, entirely bypassing digital encryption. Directing acoustic resonant frequencies at MEMS gyroscopes can induce a false angular rate reading, known as *bias injection*.

When these sensors are successfully spoofed, maliciously altered data is fed directly into the flight software. In an attempt to correct this non-existent error, the system automatically commands the physical actuators, causing the satellite to physically tumble out of control. An attacker can leverage this to drain the batteries by pointing solar arrays into deep space, or induce thermal death by rotating delicate radiators to

face the direct solar heating. These effects also show why subsystem vulnerabilities cannot be analyzed in isolation. AOCS behaviour depends on the integrity of the data it receives, the commands distributed by the onboard computer, and the internal buses that connect sensors, actuators, and flight software.

3.2 On-Board Data Handling and Internal Bus Architecture

The On-Board Data Handling (OBDH) subsystem functions as the central nervous system of the spacecraft. Internal communication buses connect all subsystems but were historically designed under the implicit assumption of a physically isolated environment, leaving them highly vulnerable to malicious lateral movement.

The **MIL-STD-1553B** bus, a military and spaceflight data standard, operates on absolute trust. It lacks inherent cryptographic authentication. The standard relies on a rudimentary single-bit parity check. Simultaneous errors, such as a 2-bit flip induced by an attacker's localized electromagnetic pulse, will bypass the error detection logic, allowing corrupted operational commands to execute unhindered.

As bandwidth demands outpaced older standards, the modern industry widely adopted the **SpaceWire** standard. While providing high-speed flexible topologies, it introduces extremely severe cybersecurity risks. Its Remote Memory Access Protocol (RMAP) allows one node to directly read from or write to the memory space of another remote node.¹ Once obtained, the attacker can write arbitrary executable code directly into the vulnerable physical memory space of the primary On-Board Computer, achieving full Remote Code Execution. Internal bus compromise is therefore not only a communication problem. It can become a software exploitation pathway, because unauthorized memory access, corrupted packets, or maliciously routed commands may directly affect the execution environment of the flight software.

¹RMAP authentication relies entirely on a single, one-byte (8-bit) Destination Key, which is trivial to brute-force.

3.3 Software Exploitation and Mathematical Manipulation

The vulnerabilities of spacecraft software are brought into reality when examining aerospace codebases. NASA's Core Flight System (cFS) is a highly successful, open-source flight software framework. Recent audits have uncovered critical software flaws, including heap-based buffer overflows in packet handlers. When the flight software parses an incoming packet, it fails to perform bounds checking, allowing an attacker to transmit an oversized payload that overwrites critical memory structures.

One of the most insidious cyber-physical attack methodologies bridges software manipulation with orbital mechanics. The AOCS relies on an Extended Kalman Filter (EKF) to fuse noisy sensor data with the satellite's rigid-body dynamics. To protect against random hardware failures, Fault Detection, Isolation, and Recovery (FDIR) algorithms monitor the measurement residual using an anomaly detector.

However, a sophisticated cyber-adversary executing a stealthy False Data Injection Attack mathematically derives a specific bias injection vector to add to the legitimate measurement. To bypass the FDIR, the attacker ensures the vector resides entirely within the column space of the observation matrix. Because the injected bias perfectly aligns with the system's own internal dynamic model, the anomaly detector sees no disruption whatsoever. The satellite unwittingly accepts the poisoned data, silently thrusting itself out of a stable orbit while downlinking falsely "nominal" telemetry.

A similar cyber-physical logic applies to the spacecraft energy system. In this case, the attacker does not necessarily need to falsify the attitude estimate or seize direct actuator control: by manipulating critical power-management decisions or forcing abnormal battery operating cycles, a cyber operation can progressively degrade the actual physical ability of the spacecraft to sustain the entire mission.

3.4 Electrical Power Subsystem Threats

The Electrical Power Subsystem (EPS) is the spacecraft's life support, managing energy generation, storage batteries and regulation. It supplies precise voltages to the On-Board Computer (OBC), payload, and attitude control. If compromised, the vehicle rapidly loses thermal regulation and communication, leading to total failure.

Unlike traditional IT attacks, EPS-oriented cyber operations do not require physical access to the hardware. Instead, a compromised OBC can indirectly sabotage the power system by modifying Battery Management System (BMS) parameters. An attacker can alter over-voltage or temperature protection thresholds, forcing lithium-ion cells to operate outside their safe design envelope. In the vacuum of space, where heat rejection relies solely on conduction and radiation, this manipulation can trigger thermal runaway, outgassing, or structural swelling of the battery cells.

A sophisticated example of this cyber-physical intersection is the **STARMELT attack**, which targets modern Satellite Internet Constellations in Low Earth Orbit (LEO). These satellites frequently enter eclipse periods where they cannot generate solar power and must rely exclusively on onboard batteries. To survive these cycles, flight software normally places critical high-consumption subsystems, like RF transponders and laser crosslinks, into deep low-power hibernation modes.

STARMELT² subverts this strategy by injecting carefully timed, low-bandwidth malicious traffic through the constellation. By forcing the victim satellite to process or route data precisely during its eclipse phase, the attacker prevents the hardware from entering hibernation. This exploits *tail energy consumption*, where transponders continue to draw significant baseline power even when processing minimal traffic, "locking" the satellite into a high-draw state while it is unable to recharge.

The impact of STARMELT is especially insidious because it is highly cumulative. Repeated over hundreds of orbital cycles, this mechanism increases the battery Depth of Discharge (DoD) and accelerates irreversible chemical degradation. There is no single *destructive command*; instead, the spacecraft's physical lifespan is effectively shortened through malicious software-driven exhaustion. This demonstrates a central pattern in modern space security: cyber compromise propagates through engineering dependencies until it produces tangible physical loss of the asset.

²STARMELT is used here as an example of a low-bandwidth, software-driven attack that accelerates battery degradation in satellite internet constellations by manipulating power-management behaviour during eclipse phases.

Transition to Systemic Infrastructure Impact

When a threat actor successfully exploits these profound technical vulnerabilities, from bypassing the FDIR with mathematical precision to draining the energy subsystem via constellation routing attacks, the impact is rarely confined to the vacuum of space. The takedown of a single critical satellite, or a cluster within an interconnected megaconstellation, immediately initiates a cascading domino effect on terrestrial infrastructures. Global banking transactions halt, precision maritime logistics lose their navigation coordinates, and national defense networks are severely blinded. This exact translation, from a localized technical cyber-physical compromise in orbit to sweeping societal and economic paralysis on Earth, will be thoroughly analyzed in *Part II: Risk Analysis and Governance*.

Part II

Risk Analysis & Governance

Chapter 4

Risk Assessment and Societal Impact

Space systems have historically relied on *security through obscurity* within an isolated engineering discipline. By shifting to Commercial Off-The-Shelf (COTS) components and LEO constellations, we have democratized space access while exposing these systems to vulnerabilities. SWaP-C limitations on onboard computers often make encryption impractical. As a result, accepting security risks ends up being a management decision rather than a technical one. This chapter explores, through the CIA triad, how space vulnerabilities cascade into catastrophic **socioeconomic failures**.

4.1 Critical Space Dependencies in Modern Infrastructure

Space assets are no longer tools reserved for scientific exploration: they are the invisible backbone of modern infrastructure. Global economies, national security and daily civilian life are linked to the uninterrupted operation of space-based systems.

Banking and Financial Systems. The global financial system operates on a foundation of absolute time synchronization, almost entirely satisfied by GPS satellite navigation. Banks, stock exchanges and international transfer networks must synchronize their internal operations through highly accurate atomic clocks onboard these satellites, within sub-millisecond tolerances. This extreme precision is a regulatory mandate in high-frequency trading, where algorithms execute transactions in fractions of a microsecond, in order to prevent market manipulation and secure cryptographic payments. Because of this deep structural dependency, the loss of verifiable satellite time can destabilize the trust mechanisms of global markets.

Global Logistics and Supply Chains. Globalization is fundamentally a maritime enterprise, relying entirely on continuous satellite connectivity to manage the vastness of international trade. Modern vessels depend on satellite navigation for collision

avoidance and precise routing, while continuous space-to-ground data exchange enables modern commerce. On land, shipping ports rely on high-precision satellites to transfer cargo from ships directly to road networks. Disrupting these digital maritime systems can compromise the global flow of materials and consumer goods.

Weather Forecasting, Agriculture and Aviation. Satellite Earth observations form modern meteorology, agricultural economics and disaster management. By continuously monitoring atmospheric conditions and soil moisture, trusted datasets drive predictive models used from insurance risk assessments to global food security optimization. In aviation, this real-time space data is a matter of life safety, allowing pilots to predict and route around severe atmospheric hazards like invisible turbulence.

Military Operations and National Security. Modern military systems operate under the assumption of uninterrupted space power. Satellite navigation enables the coordination of troop movements and the synchronization of encrypted communications. Because space is a contested operational domain, allied forces leverage military satellite networks for global control. State-sponsored adversaries actively target space infrastructures to degrade these critical capabilities during conflicts.

4.2 The Domino Effect: Cascading Failures

Because critical infrastructure sectors are deeply interdependent, cyber compromise of a single space asset triggers a cascading domino effect where the failure of one system accelerates the collapse of another, multiplying the damage exponentially.

- **Financial Desynchronization:** Signal jamming or spoofing of satellite navigation desynchronizes global master clocks. Without precise timestamps, high-frequency trading networks collapse. This represents a direct attack on data *Integrity* and *Availability*, leading to transaction mismatches, algorithmic corruption and an estimated \$1 billion daily economic impact in the U.S. alone.
- **Supply Chain Paralysis:** Navigation spoofing is a precise violation of data *Integrity*. When attackers transmit counterfeit satellite signals, onboard hardware accepts the corrupted data without triggering software alarms. This false positioning can lead directly to vessel groundings, paralyzing the supply chains.
- **Environmental Blindness:** The disruption of Earth observation telemetry drastically degrades weather prediction models. In the aviation sector, the inability to accurately track atmospheric hazards leads to catastrophic accidents. The collapse of precision agriculture models causes unmitigated crop shocks.

4.3 Real-World Case Study:

The 2022 Viasat KA-SAT Attack

The theoretical risks of space cyber threats materialized on February 24th, 2022, the day of Russia's invasion into Ukraine. A cyberattack, aiming to paralyze Ukrainian communications, disrupted satellite access with significant impact across Europe.

Attack Mechanism. Adversaries exploited a misconfigured VPN at a ground management center in Turin. This critical breach of *Confidentiality* allowed attackers to deploy *AcidRain*, a malware that overwrote the flash memory of tens of thousands modems, permanently bricking the hardware. A simultaneous denial-of-service attack flooded Viasat's management systems, achieving a total denial of *Availability*.

Unintended Domino Effect. While explicitly targeted at severing Ukrainian military communications, the attack caused massive civilian spillover. As the network collapsed, a major German energy company lost remote monitoring for 5,800 wind turbines. Simultaneously, tens of thousands of civilian subscribers across France, Italy, Poland, Greece and Hungary suffered sudden and total internet outages.

Organizational and Management Failures. The incident exposed profound flaws in the governance and architectural design of commercial space infrastructure. The technical vulnerabilities were severely aggravated by administrative blind spots:

- **Lack of Zero Trust Architecture:** A single compromised VPN granted extensive access to the trusted management network. Implementing the principle of *never trust, always verify*, would have blocked the expansion of the malware.
- **Downstream Blind Spots:** Management was unaware that critical operational technology, such as the German wind farm, relied on their broadband, highlighting a severe gap in customer risk profiling and dependency tracking.
- **Logistical Bottlenecks:** Because Viasat operated primarily as a wholesale provider without direct logistical links to end-users, distributing 30,000 replacement modems became a logistical nightmare and prolonged recovery effort.

The Viasat KA-SAT case shows that a single architectural weakness in the ground segment can generate consequences far beyond the original target. For this reason, the next step is to translate the identified vulnerabilities into measurable scenarios, assigning likelihood, impact, criticality and mitigation controls in a reproducible way.

Chapter 5

Empirical Data Analysis of Space Vulnerabilities

The preceding chapters established the architectural vulnerabilities of space systems and the cascading impacts triggered by them. This chapter translates that qualitative analysis into the technical part of the project: a Python tool for *cyber risk prioritization and treatment*. The tool implements a simple model in which each scenario is linked to one asset, one threat, some risk assessment scores and a set of security controls, converting the theoretical discussion into a measurable report.

5.1 Risk and Probability Assignments

To evaluate the threat, the assessment engine utilizes a discrete **1 to 5** scale for both *Likelihood* and *Impact*, in accordance with standard risk management frameworks.

- **Likelihood (1-5):** the probability of a successful exploit. Values are assigned based on the system's exposure and SWaP-C limitations. For instance, SpaceWire buses lacking cryptographic authentication receive high scores (4 or 5) due to their susceptibility, whereas isolated components receive lower scores.
- **Impact (1-5):** quantifies the severity of operational and societal consequences. Driven by the analysis in Chapter 4, assets that may lead to critical terrestrial desynchronization or loss of vehicle control are assigned maximum scores.

The initial risk is computed as the product of these two factors. The algorithm then applies a greedy mitigation strategy by evaluating currently deployed *Security Controls*. Each control provides a quantifiable reduction in either likelihood or impact. The program calculates the *Residual Risk* ensuring the variables never drop below 1:

$$\text{Residual Risk} = \max(1, \text{Likelihood} - \text{Reduction}_L) \times \max(1, \text{Impact} - \text{Reduction}_I)$$

If the residual risk exceeds the asset's acceptable threshold, the tool recommends additional controls until the risk is brought within acceptable regulatory limits.

5.2 Assets, Threats and Mitigation Matrix

This matrix synthesizes the attack vectors with their corresponding strategic controls.

Target Asset	Threat Scenario	Security Control (Mitigation)
GNSS Ground Stations	Lack of Zero Trust architecture and VPN misconfigurations.	Zero Trust Architecture & PSNC Mandatory Asset Auditing.
AOCS & Sensors	Physics-based analog signal spoofing bypassing digital encryption.	Human Factors Engineering & Telemetry Cross-Referencing.
Internal Space Buses (SpaceWire)	Unauthenticated lateral memory manipulation (RMAP).	Cryptographic Authentication & Privacy by Design (Edge Computing).
COTS Hardware	Exploitation of firmware backdoors via opaque supply chains.	Mandatory CVCN Hardware Screening & rigorous SCRM.
TT&C Uplink	Unauthenticated command injection and replay attacks due to legacy hardware constraints.	GDPR Art. 32 Resilience & PSNC Annex B Penetration Testing.

Table 5.1: Space Cyber-Physical Assets, Threats and Controls for JSON files

5.3 Interpreting the Tool Output

The JSON output confirms the logic of the risk model. Out of five scenarios, four are initially classified as not acceptable, while one is `already_acceptable`.

Scenario S4, related to COTS hardware supply-chain risk, already includes the Mandatory CVCN Hardware Screening and SCRM control. For this reason, the tool classifies it as `already_acceptable`. In contrast, the SpaceWire scenario reaches the acceptable threshold after the recommended cryptographic authentication control, producing a `threshold_met` result. Other scenarios remain above their thresholds even after the best control is recommended, producing a `threshold_not_met` result.

The Python tool shows that, since in space systems a single technical mitigation is not always sufficient to reduce the risk to an acceptable threshold, technical controls must be complemented by organizational governance and regulatory compliance.

Chapter 6

Governance, Policy, and Mitigations

The previous chapters showed that space cybersecurity cannot be reduced to the protection of a single satellite component, since compromise can propagate into critical services on Earth. For this reason, the final layer of defence must be organizational and regulatory, supported by governance structures aimed at long-term resilience.

6.1 Human Factors and Supply Chain Risk

The weakness in space infrastructure is not always inside the satellite. Risk often emerges from human operators, vendors and external providers that manage the ground segment and the supply chain. Operators monitor complex telemetry and manage critical commands under time pressure. In this context, fatigue and cognitive overload can increase the probability of mistakes. **Human Factors Engineering** is a necessary mitigation: operators must be trained to recognize abnormal patterns and escalate uncertain events instead of relying only on automated confidence indicators.

This is particularly relevant for AOCS and sensor-related attacks. Physics-based spoofing may bypass digital encryption because it acts directly on analog signals. For this reason, technical monitoring must be complemented by human procedures: operators should compare sensor readings with independent telemetry sources and treat inconsistent physical behaviour as a possible cyber-physical warning sign.

Supply-chain risk is a second major challenge. Modern space missions rely on COTS components and third-party ground infrastructure. This reduces cost and accelerates deployment, but introduces the possibility of limited visibility over the full lifecycle of hardware and software. Rigorous **Supply Chain Risk Management** (SCRM) is essential for components that are difficult, or even impossible, to replace.

6.2 Regulatory Frameworks

To address these organizational risks, regulatory frameworks are increasingly focused on both data protection and national security. In the context of space systems, this means that operators must protect not only the spacecraft itself, but also the data flows, ground stations, suppliers and ICT infrastructures that support the mission.

The European Union GDPR

The General Data Protection Regulation (GDPR) is relevant for satellite operators whenever their services involve the processing of personal data related to European citizens. This may include geolocation data, Earth observation imagery, broadband traffic metadata or other sensitive information generated by space-based services.

- **Article 32** requires processors to implement organizational measures appropriate to the risk. The need is to protect the confidentiality, integrity, availability and resilience of processing systems. High-impact assets require stronger controls, documented procedures and the ability to restore availability after disruption.
- **Article 33** introduces the obligation to notify personal data breaches to the competent supervisory authority without delay and, where feasible, within 72 hours. In a satellite context, this makes incident detection a requirement.
- **Article 35** requires a Data Protection Impact Assessment when a type of processing is likely to result in high risk for individuals. Large-scale satellite monitoring and geolocation services can fall within this logic when they process sensitive information. A DPIA forces the operator to identify privacy risks before deployment and to correctly document the mitigations adopted.

Finally, the GDPR principle of privacy by design and by default reinforces that protection cannot be added only after launch. In space systems, security and privacy must be integrated into the architecture from the beginning. Examples include data minimization, encryption of sensitive flows, access control on ground systems and edge-processing techniques that reduce the amount of data sent back to Earth.

The Italian National Cybersecurity Perimeter

While the GDPR mainly focuses on personal data protection, the Italian National Cybersecurity Perimeter (Perimetro di Sicurezza Nazionale Cibernetica, PSNC) focuses on the protection of networks, information systems and ICT services whose malfunctioning could damage national security.

For strategic operators, this approach is highly relevant to the space domain. Ground stations, mission control centres, communication links, and supplier-managed ICT services may all become critical assets when they support essential services such as navigation, telecommunications, defence, emergency response or Earth observation.

The PSNC introduces organizational duties that map directly to the identified risks:

- **Asset mapping and governance.** Operators must identify and maintain an updated view of the essential ICT assets. This addresses one of the main failures observed in the Viasat case: the lack of visibility over dependencies.
- **Accelerated incident reporting.** Incidents affecting critical ICT assets must be reported to the competent national structures within strict time windows. This requirement pushes operators to develop continuous monitoring, clear escalation procedures and fast technical assessment capabilities.
- **Supply-chain screening.** The involvement of the National Evaluation and Certification Center (CVCN) strengthens the control of critical ICT acquisitions. This is directly connected to the COTS hardware analyzed scenario, where supply-chain backdoors are mitigated through rigorous SCRM.
- **Testing and resilience.** The PSNC logic also supports penetration testing, infrastructure monitoring, continuity planning and disaster recovery. These measures are essential for TT&C uplinks and ground-segment services, where a compromise may rapidly become a mission-level failure.

The PSNC complements the GDPR. The GDPR protects personal data and individual rights, while the PSNC strengthens the resilience of strategic ICT systems. Together, they transform cybersecurity from a technical issue into a governance obligation.

6.3 Integrated Defence Strategy

The analysis developed in this project leads to a single conclusion: protecting space-based assets requires *more* than isolated technical controls. The vulnerabilities identified in the previous chapters affect communication links, onboard subsystems, ground infrastructure, human operators and supply chains. The defence of space infrastructure must combine engineering, risk assessment and governance.

The following table summarizes the connection between the main vulnerabilities, their potential impact and the strategic mitigations discussed in this report.

Vulnerability	Technical Cause	Potential Impact	Mitigation
GNSS Ground Stations	Weak VPNs and insufficient Zero Trust controls.	Loss of trusted timing and financial/logistical services.	PSNC mapping, Zero Trust, and continuous monitoring.
AOCS and Sensors	Analog spoofing of sensor inputs.	Loss of attitude control, thermal damage.	Human Factors Engineering and cross-referencing.
Internal Space Buses	Unauthenticated memory access and lateral movement.	Remote manipulation and mission degradation.	Cryptographic authentication and privacy-by-design.
COTS Hardware	Opaque supply chains and firmware backdoors.	Persistent compromise or loss of space assets.	CVCN hardware screening and vendor qualification.
TT&C Uplink	Replay attacks and command injection.	Unauthorized execution and loss of spacecraft control.	Penetration testing and PSNC Annex B governance.

Table 6.1: Synthesis of Space Cyber Threats, Impacts and Mitigations

The Python tool developed in the technical part supports this conclusion. It provides a reproducible way to prioritize risks and evaluate controls, but it also shows that some scenarios remain above the acceptable threshold after a single mitigation. This residual risk explains why regulatory and organizational measures are necessary.

Overall, space cybersecurity must be treated as a layered problem. Technical hardening, cryptographic authentication and monitoring reduce specific attack paths; human training, supplier assurance, incident reporting and compliance reduce the systemic consequences of those attacks. Only this combination can limit the risk that a local cyber-physical compromise in orbit becomes a critical failure on Earth.

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